

ALEX BENSON

Fort Collins, CO | (630) 841-5645 | alexander.benson@me.com | www.benson.engineer

TECHNICAL SKILLS

Languages: C, C++, Python, Java, ARM ASM, MATLAB, Visual Basic

Embedded Systems: STM32, Zephyr RTOS, QNX RTOS, SPI/I2C/CAN, PCIe, TCP/IP

Hardware: Altium, RF PCB Design, phased arrays, transmission line theory, S-parameters, RF testing

Tools & Software: MATLAB/Simulink, CAD, Git, Linux, NI LabView, oscilloscopes, signal generators, VNA

EDUCATION

Colorado State University

Master of Science, Electrical Engineering

Expected May 2027

Fort Collins, CO

- Focused on design of Embedded Systems with AI & Machine Learning.

Colorado State University

Bachelor of Science, Electrical Engineering; Concentration in Aerospace; Minor in Mathematics; 3.4/4.0

May 2026

Fort Collins, CO

- College of Engineering Dean's List (FA22, SP24, SP26)

WORK EXPERIENCE

Bureau of Reclamation, U.S. Department of the Interior

Electrical Engineering Intern; Power System Analysis & Control Group

May 2026 – Present

Lakewood, CO

- Developing custom QNX RTOS device drivers for DAQ hardware via register-level programming and PCI memory reverse engineering.
- Ownership over exciter model integration with QNX platform, interface of digital controller and HMI in VB.

FIRST RF

Embedded Systems Engineering Intern

May 2025 – Aug. 2025

Boulder, CO

- Co-developed a half-duplex phased array system for land and maritime platform connectivity.
- Designed multilayer RF PCBs in Altium with GHz transmission-line modeling and stackup optimization.
- Wrote SPI drivers in C for Zephyr RTOS to control active RF components.
- Automated passive component validation by processing S-parameter datasets in Python.

Bureau of Reclamation, U.S. Department of the Interior

Electrical Engineering Intern; Power System Analysis & Control Group

May 2024 – Feb. 2025

Lakewood, CO

- Tested hydropower governor and excitation control systems for WECC & NERC compliance.
- Designed and deployed a field transducer for signal conditioning at western U.S. power plants.
- Built embedded C firmware (QNX RTOS, STM32) for PCIe DAC/ADC control, shared memory IPC, and Simulink model injection over TCP/IP.

Woodward, Inc

Electrical Engineering Intern

May 2023 – Aug. 2023

Loves Park, IL

- Tested and evaluated data acquisition hardware for aerospace response testing.
- Developed a Python tool for large-dataset analysis and visualization, eliminating reliance on NI DIAdem.

PROJECTS

NASA University Student Launch Initiative, Payload Team Lead

Senior Design Project at Colorado State University

Fall 2025 – Spring 2026

- Managed a 9-member team to design, build, and test a deployable soil collection and pH analysis payload.
- Led full electrical design and developed autonomous firmware for payload control and data acquisition.
- NAR Certified for L1 High Power Motors.